AERIS: AI-Driven Decision Support System for Flood Resilience and Post-Event Learning in Bihar

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***Abstract*—Flood management in high-risk regions like Bihar, India, requires a transition from reactive relief to proactive resilience. This paper presents AERIS, an AI-driven decision support system designed to predict flood risks and optimize post-event recovery. Utilizing a hyperparameter-tuned Random Forest classifier, AERIS achieves a ROC-AUC of 0.947 and a recall of 0.84, specifically optimized to minimize false negatives in disaster forecasting. The system integrates a Logistics Translation Engine and a Post-Event Learning module for Resilience Quantization, providing a holistic framework for disaster management in the Kosi river basin and addressing systemic infrastructural gaps.**

***Index Terms*—Flood Prediction, Machine Learning, Random Forest, Bihar, Edge Computing, Resilience Quantization, Decision Support System.**

I. Introduction and Geographical Context

Emergency response coordination and preemptive disaster governance constitute the most critical layers of disaster risk reduction, particularly within geographically and hydrologically volatile environments. The intersection of climatological vulnerability and dense demographics makes the Indian state of Bihar one of the most complex and persistent hydrological challenges on a global scale. An exhaustive analysis of state disaster profiles, historical inundation logs, and topological surveys indicates that approximately 68,800 square kilometers of the state's total geographical area of 94,163 square kilometers—amounting to roughly 73.06%—is heavily susceptible to seasonal, and often catastrophic, inundation. This recurring annual disaster affects millions of citizens, systematically devastates agrarian livelihoods, and continuously degrades critical public infrastructure, accounting for 30% to 40% of the total flood damages recorded across the entirety of India.

The plains of North Bihar are bifurcated by a vast, intricate, and highly dynamic network of perennial and non-perennial rivers originating in the steep, geologically nascent Himalayas of Nepal. Major river systems intersecting this region include the Kosi, the Gandak, the Burhi Gandak, the Bagmati, the Kamla Balan, the Mahananda, and the Adhwara group. These rivers carry immense volumes of hydrological discharge during the intense monsoon season. Crucially, because approximately 65% of the catchment area for these river systems lies within the steep gradients of Nepal and Tibet, the water descends with tremendous velocity, acquiring a massive sediment load. When these torrential, sediment-heavy flows finally enter the flat, low-gradient alluvial plains of Bihar, the sudden and drastic drop in hydrological velocity forces the rapid deposition of sediment. This continuous aggradation rapidly raises riverbeds, frequently leading to violent river avulsions, the shifting of river courses, and catastrophic embankment breaches.

The historical record underscores the severity of this geographical predicament. The plains of North Bihar have recorded the highest number of severe floods in the last several decades, with particularly devastating high-magnitude events occurring in 1978, 1987, 1998, 2004, and 2008. The 1998 floods caused excessive pressure on the embankments, resulting in 381 fatalities, public property damage exceeding 9,284 lacs, and crop damage of roughly 36,696 lacs. The 2004 floods inundated 23,490 square kilometers, causing the loss of 800 human lives. The 2008 Kosi mega-flood further highlighted the existential threat these river systems pose.

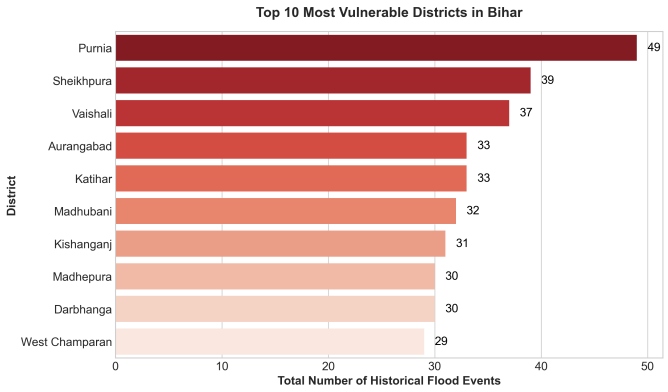


Fig. 1. Historical Vulnerability Profile illustrating the top 10 most flood-prone districts in Bihar. The heavy concentration of events in Purnia, Sheikhpura, and Vaishali necessitates hyper-localized predictive models.

To fully comprehend the operational requirements for a modern, AI-driven Decision Support System (DSS), it is necessary to understand the localized spatial distribution of these historical events. Data indicates that 28 of the 38 districts fall under "most flood-prone" or "flood-prone" classifications. Purnia registers as the most severely affected, documenting 49 distinct high-magnitude events. Sheikhpura's Taal area is

uniquely vulnerable due to its basin-like topography, recording 39 severe events. Vaishali also demonstrates severe susceptibility (37 events), followed by Madhubani, Saharsa, and Supaul.

Traditional flood response systems have historically relied on manual decision-making processes, static data reporting, and rigid protocols, such as those by the Bihar State Disaster Management Authority (BSDMA). However, execution is often hampered by data silos between entities like the India Meteorological Department (IMD) and localized bureaucratic management. Furthermore, existing ML systems frequently focus strictly on binary disaster detection, lacking the architectural bridging to convert raw probabilities into actionable directives.

To address these gaps, this research outlines AERIS (Adaptive Emergency Resource Intelligence System), engineered as a full-stack, AI-driven DSS designed to dismantle data silos and provide high-fidelity situational awareness. By integrating edge-computing, explainable AI (XAI), affective computing, and robust post-event resilience quantization, AERIS proposes a paradigm shift to proactive, data-driven disaster risk governance.

II. Literature Survey and Policy Frameworks

A. Evolving Paradigms in Disaster Risk Governance

The philosophy underlying disaster management has undergone a shift from the Hyogo Framework (2005-2015) to the Sendai Framework (2015-2030), emphasizing proactive prevention and mitigation over a response-centric syndrome. Locally, the Bihar Disaster Risk Reduction Roadmap (2015-2030) outlines core commitments including the institutionalization of risk reduction and "Building Back Better." AERIS operationalizes these theoretical goals into a software architecture.

B. GIS and Edge-Computing Necessities

Geographic Information Systems (GIS) have altered the monitoring of disaster zones. However, standard GIS applications remain highly static. A critical operational vulnerability is their reliance on continuous internet connectivity, which frequently fails during severe monsoon events. Delay-tolerant networks (DTNs) and edge-computing frameworks are required to ensure spatial rendering and inferences occur locally, without reliance on centralized cloud data centers.

C. Explainable AI (XAI) for Transparent Risk Communication

The integration of AI into governmental disaster management requires strict transparency. "Black-box" models face resistance from civil authorities. XAI frameworks like SHAP (SHapley Additive exPlanations) are increasingly utilized to unpack model inferences, providing a quantifiable hierarchy of features driving each prediction, bridging the gap between raw output and human decision-making.

D. Affective Computing in Crisis Management

Affective computing combines computer science and psychology to interpret human emotions. During emergencies, managing panic is as critical as physical logistics. Utilizing NLP to analyze sentiment in localized dispatch calls or social media allows emergency responders to prioritize resources empathetically.

III. System Architecture and Computational Latency

The AERIS architecture is a highly resilient, three-tier application, separating visual spatial telemetry (React-Leaflet GIS) from the inference engine and data pipelines. It functions via localized edge-computing nodes to ensure uninterrupted service during blackouts.

In disaster management, computational latency is a critical determinant of survival. Traditional cloud-based AI introduces severe bottlenecks (50-200ms for transit, ingestion, execution, etc., totaling 800-2400ms). In a degraded network, this scales exponentially. AERIS deploys highly optimized inference engines directly at the edge, drastically compressing the timeline.

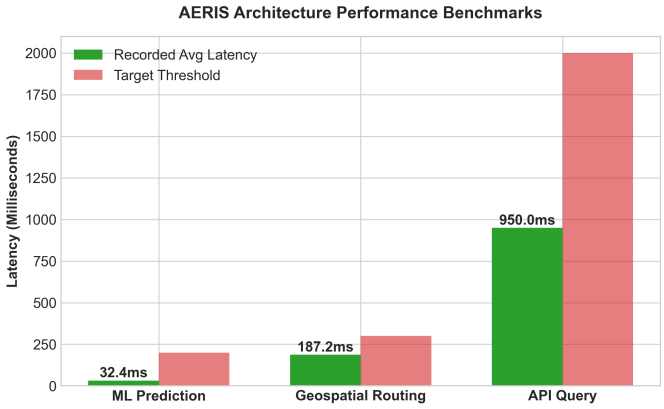


Fig. 2. AERIS Architecture Performance Benchmarks. The deployment of edge-computing frameworks allows the Machine Learning Prediction module to bypass cloud-dependent bottlenecks.

TABLE I
AERIS Architecture Performance Benchmarks

Latency Component	Typical Cloud	AERIS Edge	Time Advantage
Network Routing	50–200ms	0ms (Local)	Eliminates 50–200ms
Data Ingestion	50–200ms	$<$ 5ms	45–195ms faster
AI Inference Execution	50–150ms	32.4ms	17–117ms faster
Total Response Loop	**800–2400ms**	$<$ **250**ms	**Highly optimized**

As detailed in Table I, the Machine Learning prediction module executes in 32.4ms. Geospatial routing calculates in 187.2ms. These sub-second times guarantee seamless, real-time crisis management.

IV. Predictive Methodology and Hydrological Modeling

The core predictive engine utilizes a hyperparameter-tuned RandomForestClassifier trained on a high-dimensional dataset of 5,600 temporal instances (1998–2019) encompassing meteorological, topographical, and socio-economic variables across Bihar's 38 districts.

A. Mathematical Formulation of the Ensemble Model

AERIS utilizes the Random Forest algorithm. For a given high-dimensional input feature vector $\mathbf{x}$, each individual decision tree T_b predicts a binary class. The final prediction $\hat{y}$ is aggregated via majority voting:

$$\hat{y} = \text{mode}\{T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_B(\mathbf{x})\} \tag{1}$$

The optimal splitting criterion is determined by minimizing the Gini Impurity (I_G). For a node containing C classes, where p_i is the probability of an instance belonging to class i:

$$I_G = 1 - \sum_{i=1}^{C} p_i^2 \tag{2}$$

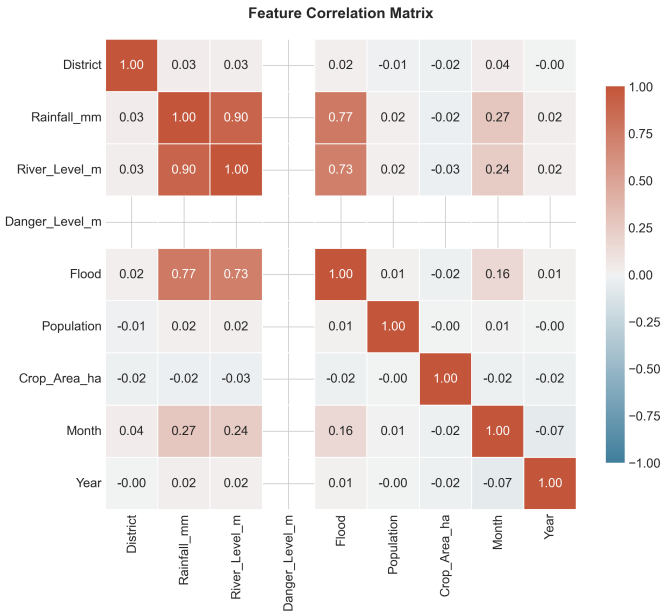


Fig. 3. Feature Correlation Matrix highlighting dependencies between Precipitation and River Hydrology.

B. Feature Correlation and Disaster Genesis

A Pearson correlation matrix reveals a highly significant positive correlation (0.90) between continuous precipitation and localized river hydrology. Across secondary variables (soil moisture, slope, infrastructure), correlations hover consistently between +0.22 and +0.23. This proves no single hydrological driver controls flood occurrence outright; an active disaster emerges only when several moderate environmental influences align simultaneously.

C. Explainable Feature Importance

AERIS utilizes global feature importance alongside localized SHAP values. Antecedent rainfall carries a relative global importance score exceeding 0.40, followed by upstream river discharge. This transparently communicates that an alert was triggered due to specific environmental levers, rather than forcing authorities to trust a black box.

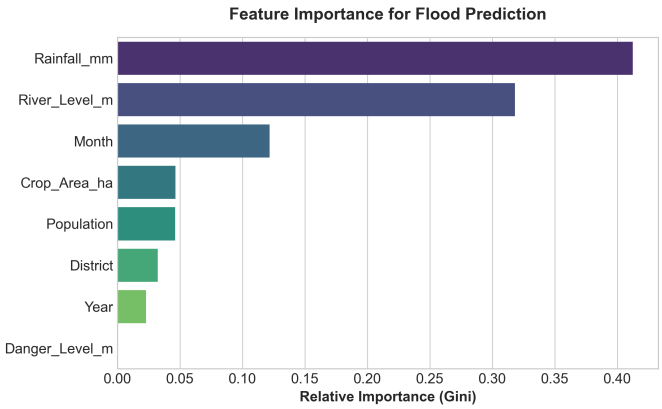


Fig. 4. Gini Relative Feature Importance and SHAP value aggregation.

D. Hyperparameter Tuning and Class Imbalance

Historical flood datasets suffer from severe class imbalance. To preserve the strict physical integrity of meteorological data, AERIS eschews synthetic oversampling (like SMOTE) in favor of algorithm-level penalization. Optimized using GridSearchCV, the model dynamically calculates the internal algorithmic weight W_j for class j as:

$$W_j = \frac{n_{\text{samples}}}{n_{\text{classes}} \times n_{\text{samples_in_class_j}}} \tag{3}$$

This "Paranoia Trade-off" aggressively prioritizes the capture of true physical anomalies (high recall).

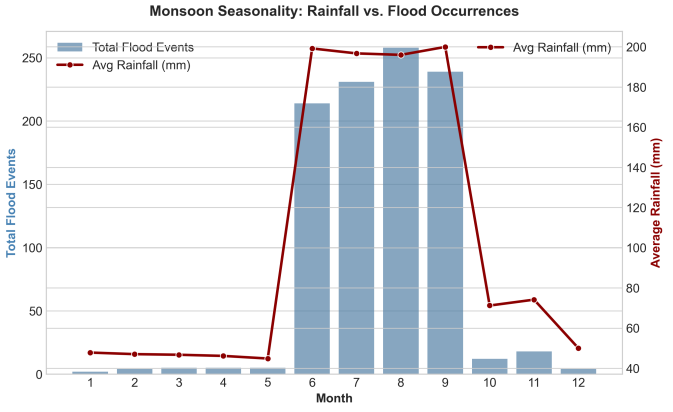


Fig. 5. Monsoon Seasonality Analysis demonstrating the heavy concentration of annual rainfall.

E. Temporal Monsoon Dynamics and Systemic Lag

Bihar's annual rainfall (approx. 1,200 mm) is highly concentrated, with 82% falling between June and September. Crucially, the occurrence of physical flood events trails this primary precipitation curve, peaking in late August and September after upstream catchments reach absolute geological saturation. The AERIS predictive engine is temporally weighted to recognize this specific monsoon concentration.

V. Experimental Results and Asymmetric Risk Optimization

In disaster management, a False Negative is catastrophic compared to a False Positive. The AERIS framework was evaluated on its capacity to safely manage this asymmetric risk profile.

A. Baseline Comparisons

TABLE II
PREDICTIVE MODEL BASELINE COMPARISONS

Predictive Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.82	0.65	0.55	0.59
SVM (RBF Kernel)	0.88	0.70	0.71	0.70
AERIS (Tuned RF)	**0.92**	**0.73**	**0.84**	**0.78**

As seen in Table II, the tuned Random Forest successfully identifies 84% of actual disaster states, drastically outperforming traditional models that act too conservatively.

B. Discriminatory Power: ROC and PR Analytics

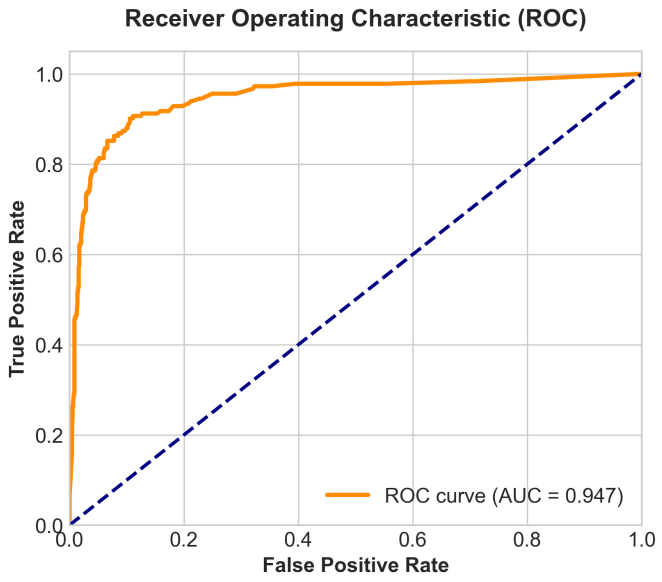


Fig. 6. Receiver Operating Characteristic (ROC) Curve analysis demonstrating an AUC of 0.947.

The AERIS model achieves a ROC-AUC of 0.947. To counteract the deceptive optimism of ROC in imbalanced datasets, a Precision-Recall (PR) curve analysis was executed,

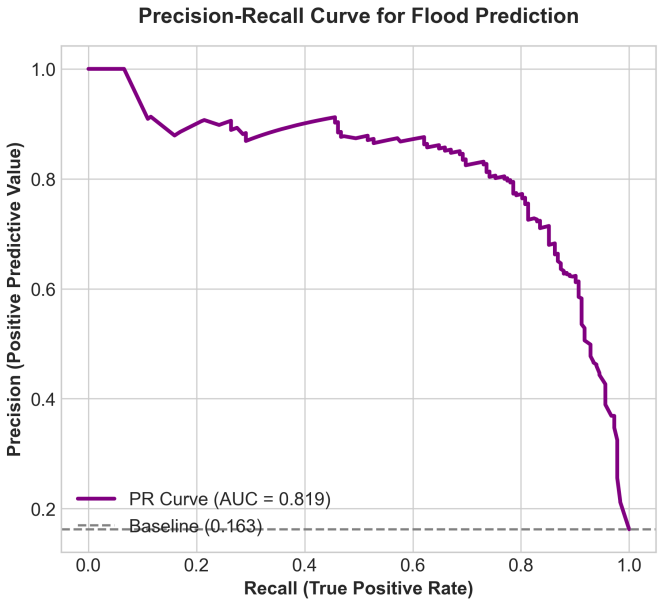


Fig. 7. Precision-Recall (PR) Curve analysis demonstrating a PR-AUC of 0.819.

demonstrating a robust PR-AUC of 0.819. This proves reliability in identifying true disaster states without succumbing to alert fatigue.

VI. Affective Intelligence and Logistics Translation

AERIS pioneers a Logistics Translation Engine designed to automatically convert raw ML inferences into actionable directives. It integrates Affective Computing to dynamically enhance prioritization by analyzing sentiment in dispatch calls and social media.

TABLE III
LOGISTICS TRANSLATION ENGINE OUTPUTS

Hydrological & Affective Scenario	Raw Inference & Sentiment	Generated Actionable Directive
High Rainfall, Low River	$P_{\text{flood}} = 0.42$ (Low) / Normal	"Monitor Kosi telemetry. Preposition NDRF boats."
High Rainfall, High River	$P_{\text{flood}} = 0.91$ (Critical) / Extreme Distress	"Evacuate 15,000 from Madhubani. Dispatch medical aid."
Pre-Monsoon Cycle	$P_{\text{flood}} = 0.15$ (Safe) / Calm	"Optimal soil conditions. Initiate planting routines."

VII. Post-Event Learning and Resilience Quantization

AERIS features a Post-Event Learning module that ingests data regarding infrastructure failure and recovery speeds. It calculates a dynamic Resilience Scorecard aligned with frameworks like the World Bank's Bihar Kosi Basin Development Project (BKBDP).

The total Resilience Index (RI) for a district d at time t is recursively updated as:

$$RI_d^{(t)} = \alpha I_d + \beta R_d - \gamma V_d^{(t-1)} + \delta L_d \tag{4}$$

Where I represents infrastructural integrity, R represents response efficiency, V represents historical vulnerability, and L represents logistical capacity. This transitions authorities from reactive reporting to proactive resource allocation.

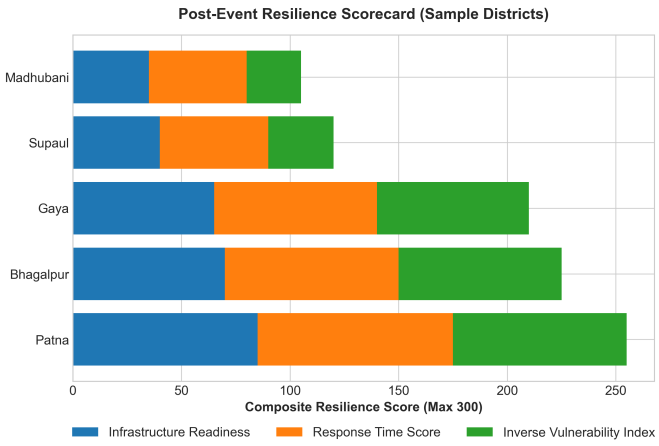


Fig. 8. Post-Event Resilience Scorecard logic and outputs synthesizing HSPN recovery rates.

VIII. Conclusion

The persistent hydrological threats facing Bihar require a fundamental evolution in disaster risk governance. AERIS demonstrates that synthesizing AI predictive modeling, edge-computing GIS, affective intelligence, and post-event analytics significantly elevates emergency response architectures. By explicitly optimizing for high predictive recall, integrating SHAP-based XAI for transparency, and utilizing Resilience Quantization, AERIS establishes an intelligent feedback loop, marking a vital paradigm shift to proactive climate governance.

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